NULL VECTORS

MEMBERS : NISHCHAY GARG
SHAURYA GOYAL
SAHIL MISHRA

PROBLEM WE SOLVE

1

Visual Analysis

Analyze camera input for drivable regions

2

Terrain Evaluation

Assess terrain difficulty and conditions

3

Navigation Commands

Convert perception into clear commands

4

Explainable Logic

Combine deep learning with rule-based logic

Unreliable Off-Road Navigation

Traditional systems fail in rough terrain

Off-Road Navigation Intelligence

Safe Off-Road Navigation

Intelligent decision-support for vehicles

Made with Napkin

KEY FEATURES

Terrain Understanding

Pixel-level detection of drivable and blocked regions using deep learning.

Hybrid Decision Intelligence

Combines Deep Learning with rule-based logic for reliable outcomes.

Image & Video Compatibility

Works with both single images and continuous video inputs.

CPU-Efficient Processing

Optimized to run without high-end GPU hardware.

Lane-Less Navigation

Generates navigation decisions without relying on roads, maps, or markings.

Explainable AI System

Transparent reasoning behind every navigation command.

Terrain Difficulty Classification

Identifies terrain as Easy, Moderate, Rough, or Very Rough.

Modular & Scalable Architecture

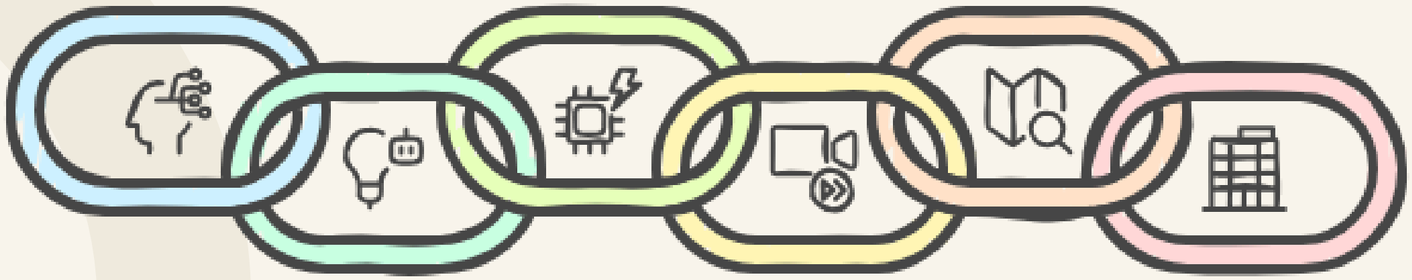
Easy integration with robotics and autonomous vehicle systems.

TARGET USERS

B2B

- Autonomous vehicle developers
- Robotics & defense organizations
- Research institutions

USP



Hybrid Intelligence
Combines deep learning with rule-based systems for robust decision-making.

Explainable AI
Ensures transparency and understanding of AI decisions.

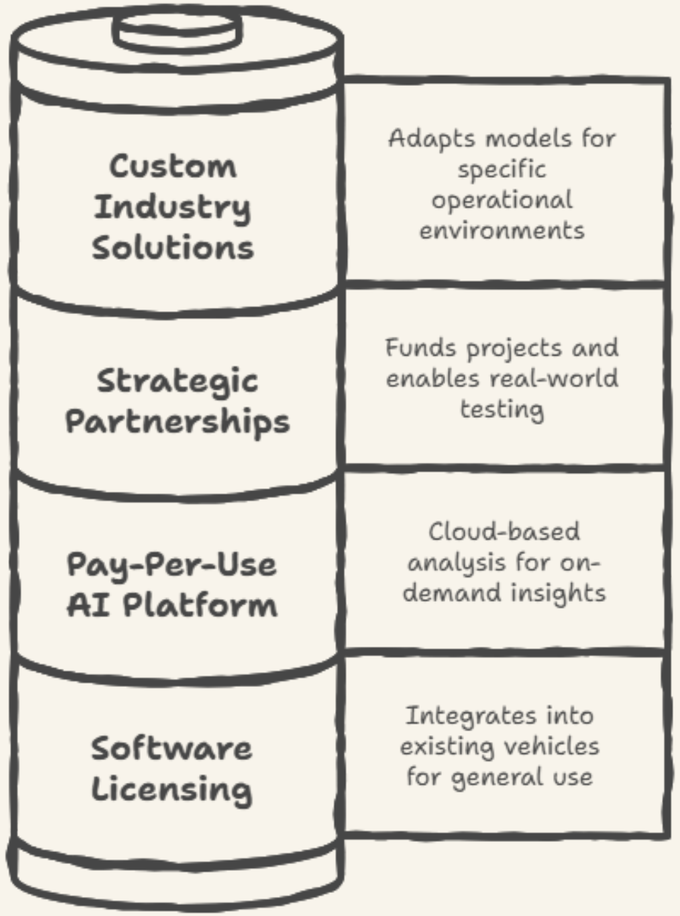
CPU-Optimized Design
Enables efficient operation without relying on GPUs.

Multi-Input Support
Handles both image and video data for versatile applications.

Zone-Based Navigation Analysis
Evaluates navigation paths from multiple perspectives.

Modular & Scalable Architecture
Allows for easy integration and expansion into robotics.

BUSNESS MODLE



FEASABILITY

Defense Applications

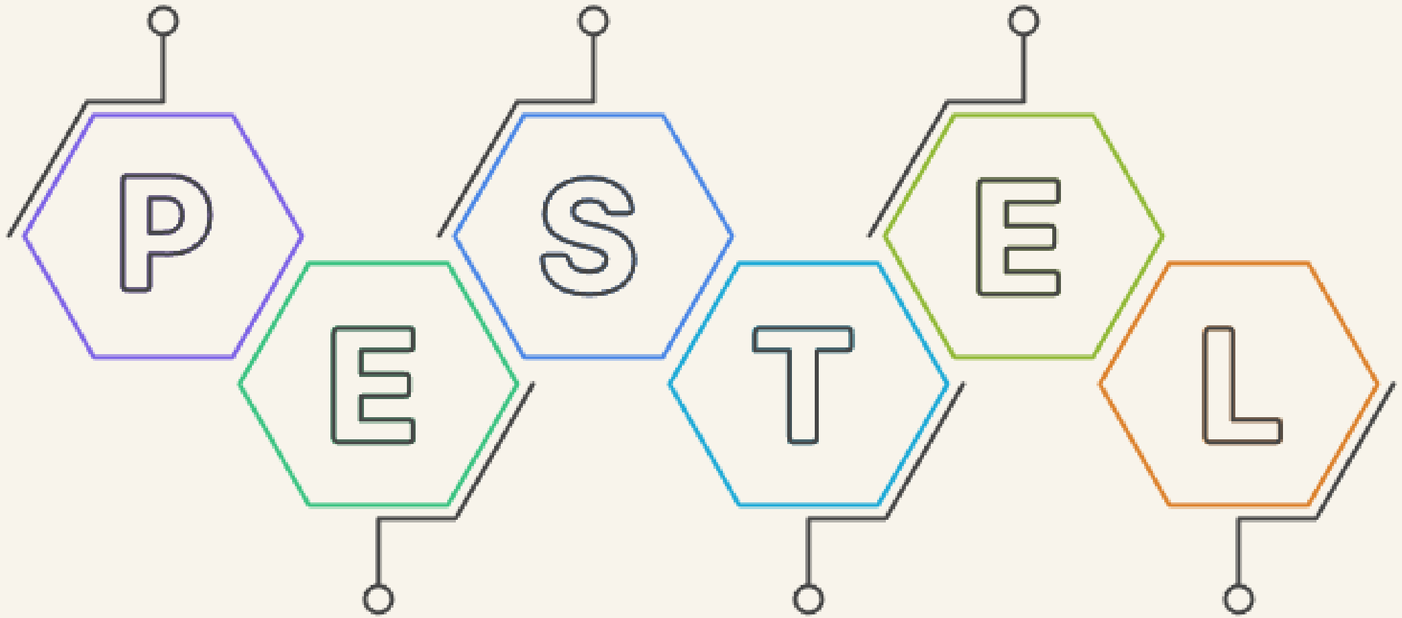
Off-road autonomy needed

User Accessibility

UI-based demo

Low-Power Platforms

Embedded systems suitable



Cost-Effective

CPU-only inference

Validated Architectures

UNet and ResNet18

No Legal Challenges

No legal challenges

SHOW STOPER

Limited Real-World Deployment

Requires domain-specific data collection

Lighting Condition Variation

Affects segmentation accuracy

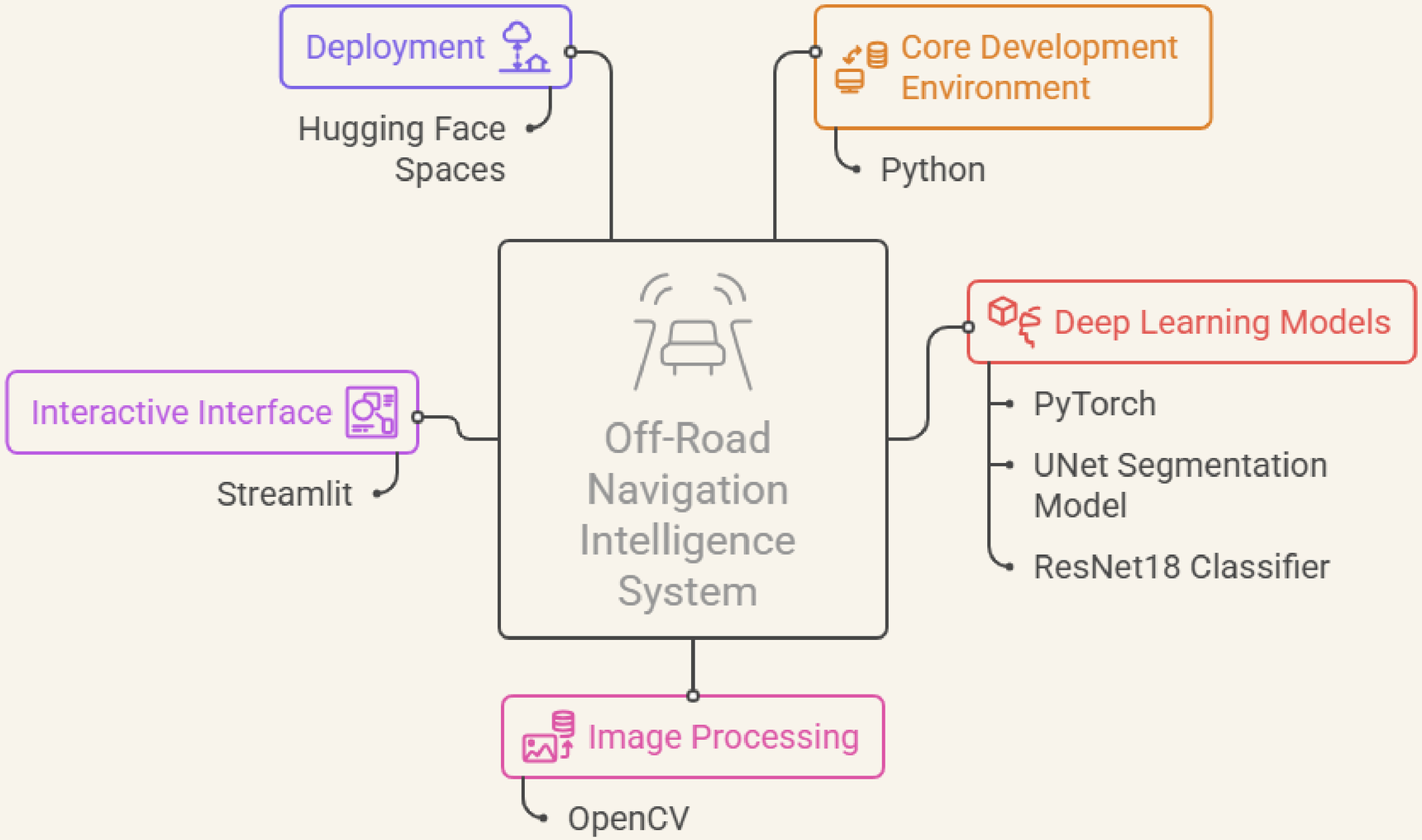
Terrain Appearance Variation

Deserts, forests, mud, rocks differ

General Dataset Reliance

Lacks real-world off-road data

TACK STACK


Deployment
Hugging Face Spaces
Core Development Environment
Python
Interactive Interface
Streamlit
Off-Road Navigation Intelligence System
Deep Learning Models
PyTorch
UNet Segmentation Model
ResNet18 Classifier
Image Processing
OpenCV

THANK YOU